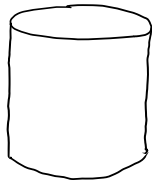


Data

How to store or handle data?

Database

container for
storing data,
fetch, add, modify

storing huge amounts of data

→ Talk to the data → use SQL (Standard Query Language)

→ Data visualization → Power BI

→ Database Management System

- Manages the database
- Handles all requests from all sources.

→ Database lives on → Server

→ Server machine never stops or shuts down.

Types of database:

→ Relational ✓

Contains tables → tables are columns and rows → tables are connected to each other.

Ex: Microsoft SQL Server, MySQL, PostgreSQL

→ Key-Value

Similar to Python dictionary
Ex: Redis, Amazon DynamoDB

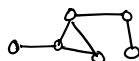
→ Column-based

Data grouped into columns.

Ex: Apache Cassandra, Amazon Redshift

→ Graph

Graphs have nodes, edges



Ex: Neo4j

→ Document

Data has no specific structure. Data is stored in a single page.

Ex: MongoDB

Query Language
Support

SQL

Relational

Non-SQL

Key-value

Column

Graph

Document

Database Structure:

Schema -

Group of related data.
(Table group)

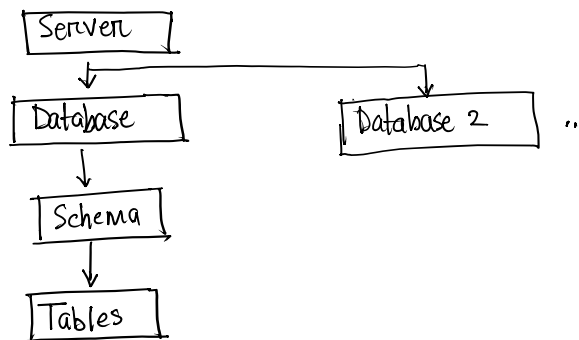
Table -

Organization of data into rows and columns.

Column (Fields) → Data type

Row (Records) → Entry of data

(Column, Row) → Gives a cell,
Contains a single value.



Datatype

- Numeric → Int, Decimal
- Text → Char, Varchar
- Date & Time → Date, Time

Types of SQL Command

CREATE [DDL]

Creates a new "Table"

ALTER [DDL]

DROP [DDL]

INSERT [DML]

Insert data inside an existing table

UPDATE [DML]

DELETE [DML]

SELECT [DQL]

Return data based on the specified Query

[DDL] — Data Definition Language (DDL)

[DML] — Data Manipulation Language (DML)

[DQL] — Data Query Language (DQL)

Why Learn SQL

- Most companies use databases to store their data. If you want to talk to their data, work there, or even store a massive amounts of data yourself, You have to learn SQL.
- Essential for data related job: Data scientist, Data Analyst, Data Engineer, Software Engineer.
- Power BI, Tableau, Spark, Kafka, Synapse
Industry standard query languages.